Round 13 - Source-Coupled Skew-Lift Quartic Carrier

ONE mechanism for a cheap carried fourth cumulant with a genuine all-distinct sector

Proposal: keep the existing source-wise third-cumulant tensors T_s exactly as the chain already does, and add only one vector r_s in R^n to each source. Represent carried fourth-cumulant memory as K4_mem = sum_s Sym4(T_s tensor r_s). This creates a genuine (1,1,1,1) sector, is closed under the same linearized ReLU/weight transport as the measured 83-90% memory carry, and adds 0 dense n x n products per layer.

1. State and exact transport

[Derived] State. For every live K3 source s, retain its existing symmetric cubic contribution T_s and attach a vector r_s in R^n. The extra memory is only the vectors {r_s}; no fourth-order tensor is materialized. With Sym4 defined as the average over the four placements,

$$K4_s[a,b,c,d] = 1/4 \; (\; T_s[a,b,c] \; r_s[d] + T_s[a,b,d] \; r_s[c] + T_s[a,c,d] \; r_s[b] + T_s[b,c,d] \; r_s[a] \;).$$

[Derived] Genuine all-distinct sector. If a,b,c,d are all distinct and T_s has all-distinct content, K4_s[a,b,c,d] is generically non-zero. Therefore this is not a diagonal or (2,2)-only carrier.

[Derived] Linear transport h_l -> z_{l+1}. With z_i = sum_a h_a W_ai, the existing source obeys T'_s = T_s x_1 W^T x_2 W^T x_3 W^T. Transport only the vector

$$r'_s = W^T \; r_s.$$

[Derived] Old-memory ReLU dressing. For D = diag(Phi(mu/sigma)), use r_s^post = D r_s^pre while the chain applies the same first-order D^{tensor 3} dressing to the old T_s source. Hence the composite carrier transforms as D^{tensor 4} K4_s. This is exactly the linearized carry tested in the prompt, but with a compressed quartic factor that remains tied to the realized K3 orientation.

2. Birth rule at a ReLU

[Derived proposal] When a new K3 source T_new is born, let d_new[i] = T_new[i,i,i]. Let q_fresh[i] be the diagonal of the chain's existing memoryless K4 birth/core at that ReLU (the same fresh object already computed today). Initialize a source vector by the separable projection

$$r_{new}[i] = q_{fresh}[i] \; d_{new}[i] \; / \; (\; d_{new}[i]^2 + \tau^2 \;),$$

[Estimated design choice] tau is only a numerical ridge, e.g. a small fraction of RMS(d_new); it is not a learned per-neuron correction. If the oracle falsifier below fails, do not tune tau - kill the representation.

[Derived] Why this is not the dead diagonal-only K4 route. The vector is never transported as a K4 diagonal. It multiplies the full carried cubic source. The resulting pair and 3+1 slices are

$$K22_s[a,b] = 1/2(\; D21_s[a,b] \; r_s[b] + D21_s[b,a] \; r_s[a] \;)$$

$$K31_s[a,b] = 1/4(\; d3_s[a] \; r_s[b] + 3 \; D21_s[a,b] \; r_s[a] \;).$$

3. Diagonal readout and mean closure

[Derived exact readout for this representation] For output column w_i of W,

$$K4_s[w_i,w_i,w_i,w_i] = T_s[w_i,w_i,w_i] \; * \; (r_s^T \; w_i).$$

Therefore, if d3'_s[i] is the per-source next-layer cubic diagonal already obtainable from the K3 source loop and u_s = W^T r_s,

$$q4_{mem}[i] = sum_s \; d3'_s[i] \; u_s[i].$$

[Derived proposal] Use q4_mem as the carried K4 diagonal input to the existing univariate mean closure. The current memoryless K4 core is used to seed r_new, not added again to q4_mem, avoiding double counting.

4. Exact incremental FLOP ledger at n = 1024

operation	real shape	exact FLOPs	u
transport all r_s	(n x n)(n x k)	2 n^2 k	k/n
ReLU slopes	n x k elementwise	n k	k/(2n^2)
quartic readout (d3'_s cached)	n x k elementwise + row sum	n k + n(k-1)	(2k-1)/(2n^2)
new-source ridge birth	length n	<= 4 n	<= 2/n^2

[Derived] Deepest layer, k = 15. The complete extra update/readout above is 31,506,432 FLOPs = 0.01467 u. Across all 15 transitions, sum k = 120, giving 252,072,960 FLOPs = 0.11738 u total.

[Derived conservative standalone count] If per-source d3 is not already cached, the simplified source formula d3_s[i] = sum_j w_j P_ij^2 A_ij costs (4 n^2 - n) FLOPs per source. Charging that again for all 120 source-uses adds 503,193,600 FLOPs = 0.23432 u. Thus the entire extra K4 mechanism is <= 0.35170 u over the network under the prompt's source formula. It uses zero dense square products.

5. Why the pattern could reach 0.7

- **[Quoted]** The exact linear K4 carry has pattern correlation 0.77-0.80 at deep layers. This carrier is explicitly closed under the same D^{tensor 4} / W^{tensor 4} linear memory transport.
- **[Derived]** Parity is correct. A K3 source is odd in later weights; r_s is transported linearly and is also odd. Their product is even, so the carrier can encode the fixed-network K4 content that even scalar summaries miss.

- **[Derived]** The dominant (2,2) sector is not discarded: K22 is generated directly from the full off-diagonal D21_s and r_s. The all-distinct 1,1,1,1 sector is also non-zero by construction.
- **[Quoted + inference]** The measured K4 memory horizon is >= 6 layers. Source vectors live exactly as long as the existing K3 sources, so the mechanism has no short-horizon cutoff. This is a structural reason for plausibility, not a measured guarantee.
- **[Estimated hypothesis]** Pattern correlation >= 0.7 is plausible only if the true K4 memory lies close to the source-coupled subspace span{Sym4(T_s tensor r_s)}. The falsifier below tests that directly before any end-to-end implementation.

6. ONE falsifier: teacher-forced source-vector oracle

[Derived test; no free-run claim] At each stored deep post-ReLU capture l in {11,14,15}, reconstruct every source's D21_s and d3_s. Using the teacher post K22 and K31 at that same layer, solve only offline for the best vectors r_s in the linear model

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min_R ||K22_hat(R)-K22_teacher||_F^2 + ||K31_hat(R)-K31_teacher||_F^2,
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with K22_hat and K31_hat given by the equations in section 2. Do not use the next layer's K4 diagonal in the fit. Then transport each fitted r_s once with the actual W_l and predict the next pre-activation diagonal by q4_hat[i] = sum_s d3'_{s}[i] (W_l^T r_s)[i].

[Quoted gate] On the decisive neurons |mu/sigma| < 1.5, require pattern correlation >= 0.70 against the teacher at the available deep one-step targets L12, L15 and L16 on both networks. If any network is consistently below 0.70, kill this representation; tuning the birth ridge cannot rescue a failed oracle subspace.

Bottom line: Source-Coupled Skew-Lift is a genuine carried K4 object, not a fresh predictor. It inherits long source memory, has a real all-distinct sector, preserves the required weight parity, and costs < 0.36 u total beyond the existing K3 chain under a conservative count. The decisive uncertainty is representational, not computational; the teacher-forced source-vector oracle should be run before any free-run implementation.